FINAL PROJECT REPORT: IMPROVED DETECTION OF FRAUD & CREDIT RISK

Project Title: BNPL Credit Risk Scoring and Fraud Detection for Bati Bank

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Target Audience: Executive Leadership, Risk Management Department, and Technical Stakeholders

1. Understanding and Defining the Business Objective

The primary objective of this project was to enable **Bati Bank** to safely launch a **Buy-Now-Pay-Later (BNPL)** service. Because the bank lacked historical default data for these specific transactions, the project aimed to:

- **Create a Credit Risk Proxy:** Establish a behavioral baseline for "High Risk" vs. "Low Risk" using alternative transaction data.
- **Improve Detection:** Move beyond simple rules-based filtering to advanced Machine Learning that can distinguish between stable customers and potential defaults or fraudulent actors.
- **Regulatory Alignment:** Develop models that provide a balance between high-performance "Black-Box" models and "Interpretable" models compliant with **Basel II** standards.

2. Discussion of Completed Work and Analysis

Data Engineering & Exploratory Data Analysis (EDA)

We processed over 95,000 transactions with **zero data loss**. Key patterns identified include:

- **Recency & Frequency:** High-risk behavior was mathematically linked to low transaction frequency combined with high recency (long gaps since last activity).
- **Monetary Skewness:** Value distributions were highly skewed, requiring **Logarithmic Transformations** to ensure outliers didn't bias the model.
- **Clustering:** Used **K-Means Clustering** to segment customers into three distinct risk categories, which served as the "Ground Truth" for our supervised learning.

Model Development & Comparison

We implemented a **Champion-Challenger** framework to prove "Improved Detection":

Metric	Logistic Regression (Baseline)	Random Forest (Improved)
Model Type	Linear / Interpretable	Non-Linear / Complex
AUC-ROC Score	[Insert LR Score, e.g., 0.72]	[Insert RF Score, e.g., 0.89]
Primary Strength	Regulatory Compliance	Superior Detection Accuracy

3. Business Recommendations and Strategic Insights

Based on the modeling results, we recommend the following strategies for Bati Bank:

- **Tiered Credit Limits:** Implement the **Random Forest** model as a primary filter. Customers flagged as "High Risk" should be limited to small, essential airtime purchases before being granted larger financial service limits.
- **Focus on Recency:** Our **Feature Importance** analysis showed that the time elapsed since the last transaction is the strongest indicator of risk. Bati Bank should trigger automated re-engagement or risk-reassessment if a customer's Recency exceeds a specific threshold.
- **Hybrid Approval Flow:** Use **Logistic Regression** for the official "Explanation" to the customer (if a loan is denied) and **Random Forest** for internal risk monitoring.

4. Limitations and Future Work

Limitations

- **Proxy Bias:** The model is trained on a "Proxy" of risk, not actual historical defaults. While statistically sound, the proxy may not capture 100% of true default behavior.
- **Static Data:** The current model uses a snapshot. It does not yet account for real-time macroeconomic shifts (e.g., inflation or seasonal spending changes).

Future Work

- **Inclusion of External Data:** Integrating mobile device metadata or social behavior could further sharpen the "Improved Detection."
- **SMOTE Implementation:** Future iterations will use **Synthetic Minority Over-sampling Technique (SMOTE)** to better handle the imbalance between high-risk and low-risk populations.

- **Real-time API:** Transitioning this Python model into a real-time API for instant credit decisions at the point of sale.

5. Report Structure, Clarity, and Presentation

This submission represents a complete end-to-end Data Science pipeline.

- **Reproducibility:** All code is modularized into `eda.ipynb` and `modeling.ipynb`.
- **Transparency:** The logic used to define "Risk" is documented via K-Means and RFM analysis.
- **Visual Evidence:** Use of **Confusion Matrices** and **ROC Curves** provides mathematical proof of the model's reliability.

Final Conclusion: By adopting the Machine Learning framework developed in this project, Bati Bank can successfully mitigate risk in its BNPL venture, ensuring that credit is extended to those most likely to repay while aggressively detecting and filtering high-risk fraudulent behavior.